

CVS · DAY 6 · HEART FAILURE & SHOCK — CLINICAL MANAGEMENT

USMLE Step 1 · June Cohort

PREPARED BY ALLAN BAKESIGA · USMLE STEP 1 PREP

bakesiga.github.io/usmle-dashboard

Print and stick on your wall · All wording original · Builds on Day 3 physiology

HF CLINICAL PRESENTATION

LEFT VS RIGHT

- **HFrEF (left):** orthopnea, PND, exertional dyspnea, fatigue, **S3**, bibasilar crackles, displaced apical impulse (inferolateral PMI).
- **Right HF / cor pulmonale:** JVD, hepatomegaly, peripheral pitting edema, **cardiac ascites** (transudate, **SAAG > 1.1**), Kussmaul sign.
- #1 cause of right HF = **left HF**. #2 = chronic lung disease (cor pulmonale).
- **Pulmonary edema histology:** pink acellular intra-alveolar transudate; chronic LV failure → **hemosiderin-laden alveolar macrophages** ("HF cells", Prussian blue +).
- **Lymphatic buffer:** ~10–20× normal capacity must be saturated before clinical edema → explains why early ↑ filling pressures may be silent.

S3 VS S4 QUICK-REF

VOLUME VS PRESSURE

	S3	S4
Timing	Just after S2 (early diastole)	Just before S1 (late diastole)
Loaded with	Volume — floppy LV	Pressure — stiff LV
Mechanism	Rapid early-diastolic filling deceleration	Atrial kick hits a non-compliant wall
Seen in	HFrEF, MR, AR, high-output, normal in < 40 y	HTN/concentric LVH, HCM, restrictive CMP, acute MI

HFREF VS HFPEF

EF · LV GEOMETRY

	HFrEF	HFpEF
EF	≤ 40%	≥ 50%
LV	Dilated, eccentric hypertrophy	Non-dilated, concentric LVH
Driver	Ischemic/DCM — ↓ contractility	Chronic HTN — diastolic dysfunction (↑ afterload)
PV loop	ESPVR slope ↓; loop shifts right	EDPVR shifts up-left (stiff)
Mortality Rx	4-pillar (β + ARNi + MRA + SGLT2-i)	SGLT2-i + MRA + BP control

ACC/AHA STAGE VS NYHA CLASS

STRUCTURE VS SYMPTOMS

ACC/AHA	What it means
A	At risk (HTN, DM, CAD) — no structural disease
B	Structural disease, asymptomatic → start ACE-i + β-blocker NOW
C	Symptomatic HF → 4-pillar GDMT
D	Refractory → LVAD, transplant, hospice

NYHA	Functional capacity
I	Asymptomatic with ordinary activity
II	Symptoms with ordinary activity
III	Symptoms with less-than-ordinary activity
IV	Symptoms at rest

ADHF CHEST X-RAY

PULMONARY EDEMA SIGNS

- Bilateral patchy **airspace opacities** (alveolar edema).
- **Kerley B lines** — short horizontal interlobular septal thickening at lung bases.
- **Bat-wing perihilar opacification**.
- Fissure thickening (sub-pleural fluid).
- Blunted **costophrenic angles** (pleural effusions).
- **Cardiomegaly** + cephalization of pulmonary vessels.
- Cardiac/vascular curves: cardiac shifts **down-right** (↓ contractility) + vascular shifts **right** (volume retention) → new intersection at **↑ RA pressure** with maintained/slightly ↓ CO.

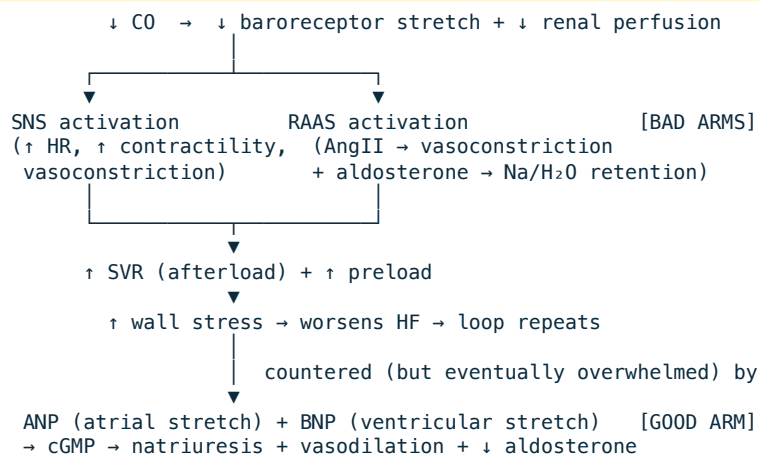
BNP BIOMARKER

WALL STRESS · NPV

- **BNP** — released by **ventricular cardiomyocytes** in response to wall stress (Laplace: $T = Pr/2h$).
- **ANP** — from atrial cardiomyocytes in response to atrial stretch.
- Both → guanylate cyclase → ↑ cGMP → natriuresis + vasodilation + ↓ aldosterone + ↓ renin.
- **Clinical use:** normal BNP *rules out* acute HF (high NPV); elevated supports HF.
- **Non-specific:** also ↑ in AFib, CKD, PE, sepsis, advanced age, female sex.
- **Falsely low** in obesity (adipose neprilysin clears BNP).
- On ARNi, BNP is unreliable for monitoring → use **NT-proBNP**.

VICIOUS CYCLE OF DECOMPENSATED HF

BAD ARMS VS GOOD ARM



- **Stage B clue:** all three neurohormones (AngII, NE, ANP) are already ↑ even before symptoms appear → rationale for starting ACE-i + β-blocker at Stage B.
- **4-pillar logic:** β-blocker breaks the SNS arm; ACE-i/ARB/ARNi breaks RAAS (ARNi also augments ANP/BNP via neprilysin block); MRA blocks aldosterone; SGLT2-i adds an independent mortality benefit.

HFREF FOUR-PILLAR REGIMEN

MORTALITY BENEFIT · START AT STAGE C (β-BLOCKER + ACE-I AT STAGE B)

Pillar	Examples	Mechanism / pearl
1. β-blocker	Carvedilol, metoprolol succinate, bisoprolol	Blocks SNS arm; reverses remodeling. Start LOW, GO SLOW. Do not start in acute decompensation.
2. ACE-i / ARB / ARNi	Lisinopril · losartan · sacubitril/valsartan	Blocks RAAS. ARNi superior to ACE-i in stable HFREF (PARADIGM-HF). ACE-i → dry cough + angioedema (bradykinin); ARB doesn't. Wash out ACE-i 36 h before starting ARNi.
3. MRA	Spironolactone · eplerenone	Blocks aldosterone → anti-fibrotic + K ⁺ /Mg-sparing. Monitor K ⁺ + Cr. Spiro = gynecomastia; eplerenone selective.
4. SGLT2 inhibitor	Dapagliflozin · empagliflozin	Works regardless of diabetes ; mechanism not fully understood. Reduces HF hospitalization + CV mortality.
Symptom add-on: loop diuretic (furosemide) — congestion relief, <i>no mortality benefit</i> .		
Devices: ICD if EF ≤ 35 % after ≥ 3 mo of optimized GDMT · CRT if QRS ≥ 150 ms + LBBB · LVAD / transplant at Stage D.		

HFPEF MANAGEMENT

STIFF LV

- **SGLT2-i** (only class with consistent mortality/hospitalization benefit in HFpEF).
- **MRA** (spironolactone) — symptom + hospitalization benefit, especially EF 45–55 %.
- **Loop diuretic** for congestion.
- **Aggressive BP control** — HTN is the dominant driver.
- Treat comorbidities: **AF, OSA, obesity, CKD**.
- β-blockers + ACE-i/ARB are NOT mortality-proven in HFpEF (used for comorbidities).

RIGHT HF · COR PULMONALE

RV PRESSURE OVERLOAD

- **COPD path:** chronic alveolar hypoxia → hypoxic pulmonary vasoconstriction + remodeling → **pulmonary HTN** → RV pressure overload → RV dilation → tricuspid regurgitation → RV failure.
- **Cardiac ascites:** ↑ portal capillary hydrostatic pressure from systemic venous congestion + normal oncotic + normal permeability → **transudate, SAAG > 1.1 g/dL**.
- Distinguish from cirrhotic ascites: both have SAAG > 1.1, but cardiac ascites total protein > 2.5 g/dL (sinusoidal congestion preserves protein leak).
- **Reversible DCMs:** alcohol (abstinence) · peripartum (often resolves) · tachycardia-induced · stress (Takotsubo).
- **Anthracycline (doxorubicin) cardiotoxicity:** dose-dependent **irreversible** DCM presenting years–decades later. Mechanism: free-radical mitochondrial damage + topoisomerase IIβ inhibition. Prevent with **dexrazoxane**.

4-TYPE SHOCK — CATH PROFILE + FIRST-LINE TREATMENT

CVP · PCWP · CI · SVR · SVO₂

Type	CVP	PCWP	CI	SVR	SvO ₂	Skin	First-line treatment
Hypovolemic	↓	↓	↓	↑	↓	Cold	IV crystalloid + source control
Cardiogenic (MI)	↑	↑	↓	↑	↓	Cold	Revascularize (PCI) + dobutamine ; IABP / Impella / ECMO
Obstructive (PE)	↑	↓	↓	↑	↓	Cold	Anticoagulate; thrombolytics if unstable; embolectomy
Septic (warm)	↓	↓	↑	↓	↑	Warm	Early antibiotics + crystalloid + norepinephrine
Anaphylactic	↓	↓	↑	↓	↑	Warm	IM epinephrine ($\alpha_1 + \beta_1 + \beta_2$ + mast cell stabilization)

- **Massive PE pathophysiology:** clot in PA → ↑ RV afterload → RV dilation + ↑ wall tension → RV ischemia (supply-demand mismatch) → ↓ LV preload → cardiogenic collapse. Echo: enlarged RV cavity, septal flattening.
- **Septic cold phase:** CO drops, SvO₂ falls, SVR may rise → looks like cardiogenic shock — context matters.
- **Tamponade quirk:** PCWP often looks ↑ (transmitted pericardial pressure), but LV is actually under-filled.

VASOPRESSOR / INOTROPE
CHEAT-SHEETMATCH TO
SHOCK

- **Norepinephrine** ($\alpha_1 \gg \beta_1$) — **first-line in septic shock** and most undifferentiated shock with low SVR.
- **Epinephrine** ($\alpha_1 + \beta_1 + \beta_2$) — **IM in anaphylaxis**; IV in cardiac arrest + refractory shock.
- **Dobutamine** ($\beta_1 \gg \beta_2$) — inotrope of choice in **cardiogenic shock** with adequate BP.
- **Phenylephrine** (pure α_1) — neurogenic shock, anesthesia-induced hypotension; reflex bradycardia.
- **Vasopressin** — adjunct in septic shock when norepinephrine alone insufficient.
- **Dopamine** (dose-dependent) — largely fallen out of favor (arrhythmias).

HIGH-OUTPUT HF PEARL

LOW SVR + ↑ CO

- Etiologies: **hyperthyroidism (Graves)**, severe anemia, large AV fistula, wet beriberi (thiamine deficiency), Paget disease of bone, pregnancy.
- Findings: **↓ SVR + ↑ CO + wide pulse pressure** + warm extremities + pitting edema + S3.
- **Graves treatment:** **propranolol** (blocks β + peripheral T4→T3 conversion) + definitive Rx (methimazole, radioactive iodine, thyroidectomy).
- Treat the cause — diuretics + standard HF meds alone do not fix it.

TOP 15 TRAPS

1. S3 = floppy LV (volume); S4 = stiff LV (pressure).
2. HFrEF ≤ 40 %; HFpEF ≥ 50 %; 41–49 % = HFmrEF.
3. Start ACE-i + β -blocker at — don't wait for symptoms.
4. 4 pillars: β -blocker + ARNi (or ACE-i/ARB) + MRA + SGLT2-i.
5. Loop diuretic = symptoms only, NO mortality benefit.
6. ICD if EF ≤ 35 % after optimized GDMT; CRT if QRS ≥ 150 ms + LBBB.
7. ACE-i cough/angioedema = bradykinin; switch to ARB.
8. ARNi: wash out ACE-i 36 h to avoid angioedema.
9. SGLT2-i works even without diabetes.
10. Septic shock first-line vasopressor = .
11. Anaphylaxis = , not IV (unless arrest).
12. Cardiogenic shock inotrope = dobutamine; revascularize early.
13. Massive PE → ↑ CVP + ↓ PCWP + RV dilation on echo.
14. Cardiac ascites: SAAG > 1.1, total protein > 2.5 g/dL.
15. Hemosiderin-laden macrophages = chronic LV failure → Prussian blue +.

© Allan Bakesiga · USMLE Step 1 Prep · Original wording. Single page — carry it and review before every Q-bank block.